



INTERNSHIP PRESENTATION

Machine Learning Applications in Intelligent Toll Management Systems

KENT Intelligent Transportation Systems (India) Pvt. Ltd.

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IILM University, Greater Noida • Internship: 15 July – 15 August 2026

Internship at a Glance

From Python fundamentals to transportation-focused machine-learning workflows

01 MONTH

Internship duration

08 TASKS

Core ML learning tasks

03 USE CASES

Classification • Prediction • Anomaly

2026

Academic batch 2026–27

Learning Focus

The internship followed a progression through data understanding, preprocessing, EDA, model development, evaluation and documentation. The transportation domain was used

KENT & Intelligent Toll Automation

Connecting machine-learning concepts with the transportation systems described in the report

Toll Automation

Vehicle detection, classification, transaction processing

Toll Xplore

Toll collection platform described as supporting trans

Traffic Management

Road, traffic and weather information supported by counters,

Vehicle / Sensor Data

AVC / ETC / Lane Systems

Data & Analytics

ML Support

Operational Review

Problem Statement & Objectives

Turning continuous transportation data into structured, testable ML workflows

Problem

Toll and transportation systems generate information about vehicle categories, la

Objectives

- Understand ML fundamentals
- Build practical Python/data-analysis skills
- Clean and preprocess data
- Perform EDA and visualization
- Study classification, regression and anomaly detection
- Evaluate and compare models
- Connect ML with intelligent toll systems

system events.

Machine Learning Workflow

A complete problem-to-evaluation pipeline



Supervised Learning

Vehicle classification → categorical target

Regression

Traffic volume → numerical target

Unsupervised

Anomaly detection → unusual observations

Technologies & Tools

The Python ecosystem used as the foundation for the learning workflow

Python

Primary programming language

NumPy

Numerical arrays & operations

Pandas

Tabular data handling

Matplotlib

Data visualization

Scikit-learn

Classical ML & evaluation

Jupyter / VS Code

Interactive & structured development

Git + Documentation

Reproducibility & organization

Data Understanding & Collection

Transportation observations represented as structured machine-learning features

Vehicle Features

Vehicle_Length
Vehicle_Height
Axle_Count
Lane_ID
Vehicle_Class

Context Features

Hour
Traffic_Volume
Payment_Mode
Day / time context
Weather variables where available

Data Quality Checks

Missing values
Duplicate records
Invalid measurements
Inconsistent categories
Incorrect data types

Preprocessing & Exploratory Data Analysis

Clean first. Understand patterns second. Model third.

Preprocessing

- Missing-value handling
- Duplicate / invalid-record checks
- One-hot encoding for categorical fields
- Scaling where required
- Avoiding data leakage by fitting transformations on training data

EDA

- Vehicle-category distribution
- Traffic volume by hour
- Numerical summaries
- Correlation analysis
- Identify patterns, imbalance and unusual observations before model selection

Vehicle Classification

A supervised-learning exercise aligned with vehicle-profile characteristics



Why it matters

Vehicle class can influence toll charging and operation

Algorithms

Decision Tree is easy to explain; Random Forest provides better accuracy

Evaluation

Accuracy, precision, recall, F1-score and confusion matrix.

Traffic Volume Prediction

Regression for estimating vehicle volume during a selected time interval

Inputs

Hour of day
Day of week
Lane identifier
Historical traffic volume
Holiday indicator
Weather variables where available

Baseline Model

Linear Regression

A simple, interpretable baseline used to understand the

Metrics & Relevance

MAE • RMSE • R^2

Potential uses: queue management, staffing, lane planning

Anomaly Detection

Finding observations that differ substantially from expected transportation patterns

Examples

Unexpected traffic volume
Unusual transaction timing
Sensor measurements outside normal ranges

Method

Isolation Forest
Unsupervised method using random partitioning; easily

Interpretation

An anomaly is a review signal — not proof of fraud, wrongdoing, or a security breach.

Model Evaluation & Comparison

Choosing models requires more than looking at one score

Classification

Accuracy
Precision
Recall
F1-score
Confusion matrix

Regression

MAE
MSE
RMSE
 R^2

Model Selection

Logistic Regression — simple
Decision Tree — explainable
Random Forest — robust ensemble
KNN — simple, scale-sensitive

Integrated ML Architecture for Toll Systems

Conceptual architecture connecting sensors, processing and ML outputs



Human Oversight

ML outputs should support operational decisions and be monitored and verified.

System Perspective

Different models serve different tasks: classification, regression and anomaly detection.

Implementation Approach

A reproducible workflow for academic ML experiments

Dataset Preparation

Separate input features X and target y .

Example:

X = vehicle length, height, axle count

y = vehicle class

Train / Test Split

20% test split

`random_state = 42`

stratification for classification where appropriate

Transformations learned from training data.

Reproducibility

Record dataset version, features, preprocessing, model con

Demonstration Evidence

Reference screenshots from the report — synthetic DEMO DATASET only

01 — Environment Setup

```
Python 3.8 environment verified  
> import pandas as pd  
> import numpy as np  
> import matplotlib.pyplot as plt  
> from sklearn.model_selection import train_test_split  
> from sklearn.metrics import accuracy_score  
All required ML libraries imported successfully
```

(DEMO DATASET — Synthetic Evidence only)

02 — Dataset Loading

```
> df = pd.read_csv('TrafficData.csv')  
> df.head()
```

Vehicle_Length	Vehicle_Height	Axle_Count	Lane_ID	Hour	Vehicle_Class
10.57	3.33	3	13	29	Bus
4.15	1.5	2	13	14	Car

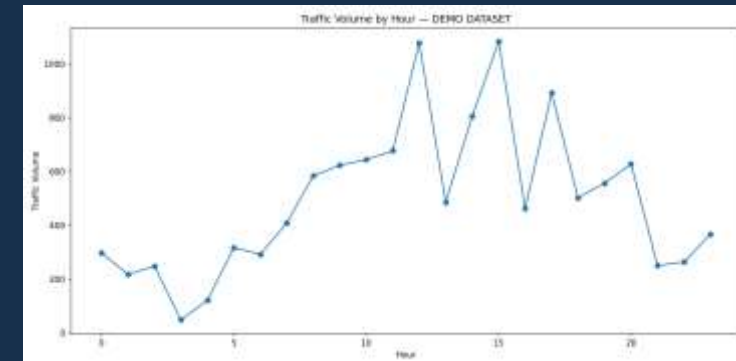
Rows: 180 (Columns: 6)

(DEMO DATASET — Synthetic Evidence only)

03 — Data Preprocessing

```
> df.isnull().sum()  
Missing values: 0 across selected fields  
> df = df.drop_duplicates()  
> OneHotEncoder(handle_unknown='ignore')  
> Standardization applied where algorithm requires scaling  
Preprocessing complete: clean, model-ready feature matrix prepared.
```

(DEMO DATASET — Synthetic Evidence only)



Environment Setup • Dataset Loading • Preprocessing • Exploratory Analysis

Testing, Results & Limitations

What the internship work demonstrates — and what it deliberately does not claim

What was achieved

- Understanding of the complete ML workflow
- Practical data-handling and modelling concepts
- Domain connection to toll automation and ITS
- Model-evaluation concepts and documentation
- Reproducible experiment structure

Important limitation

The report does not provide proprietary KENT datasets or internship experiment logs. Num

Learning Outcomes & Future Scope

From foundational ML skills toward advanced intelligent transportation applications

Technical Growth

- Python & tabular data
- Preprocessing
- EDA & visualization
- Classification
- Regression
- Anomaly detection
- Evaluation

Professional Growth

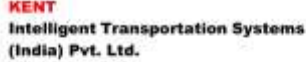
- Documentation
- Debugging
- Problem decomposition
- Reproducibility
- Technical communication
- Data-quality awareness

Future Scope

- Computer vision
- Edge AI
- Explainable AI
- Time-series / deep learning
- Real-time streams
- Monitoring & dashboards

Internship Completion Certificate

Issued by KENT Intelligent Transportation Systems (India) Pvt. Ltd.




Before You Think, IT'S Done!!

Date: 29/08/2026
Ref: KENT/HR/250826

TO WHOMSOEVER IT MAY CONCERN

It is certified that **Mr. Dev Singh** was associated with us from **15th July 2026 to 15th August 2026**. During this period, he was involved in learning and contributing to activities related to **Machine Learning** as part of his professional engagement with the organization.

We found **Mr. Dev Singh** sincere, hardworking and committed towards his assigned responsibilities. His conduct and performance during the period of his association were found to be satisfactory.

We wish him all the best in his future endeavors.

Yours Truly,

For KENT Intelligent Transportation Systems (India) Pvt. Ltd.




AUTHORIZED SIGNATORY

Dated 29th August 2026

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CONCLUSION

The internship provided a structured foundation in Machine Learning while connecting data-driven m

The key learning was the complete workflow: understand the problem → prepare data → explore → model → evaluat

THANK YOU

Questions & Discussion